SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninfor
solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

Duration : 1 Hour

QUESTIONS: 5 **Total Marks:50**

Annexure A

Analytical Problem Solving

Code of Conduct:

*I **N.SANDIYANITTIANANDAM(192571029)(**Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
SANDIYA

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

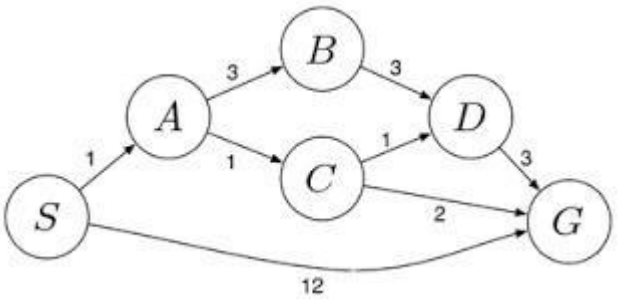
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent? ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1. Water Jug Problem

Problem: Given a 4-gallon jug and a 3-gallon jug without any measuring marks, obtain exactly 2 gallons of water in the 4-gallon jug.

Solution:

Let the state be represented as (x, y), where x is the amount of water in the 4-gallon jug and y is the amount of water in the 3-gallon jug.

Initial State: (0, 0)

Goal State: (2, 0)

Steps:

1. Fill the 3-gallon jug completely. State: (0, 3).
2. Pour water from the 3-gallon jug into the 4-gallon jug. State: (3, 0).
3. Fill the 3-gallon jug again. State: (3, 3).
4. Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full. State: (4, 2).
5. Empty the 4-gallon jug onto the ground. State: (0, 2).
6. Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug. State: (2, 0).

Result: Exactly 2 gallons of water are obtained in the 4-gallon jug.

2. Mars Rover as an Intelligent Agent

i. What are the percepts of this agent?

The Mars Rover receives information from its surroundings through sensors and cameras.

- Images of rocks, soil and terrain.
- Distance from obstacles.
- Temperature and atmospheric conditions.
- Chemical composition of soil and rock samples.
- Battery level and equipment status. **ii. Characterize the operating environment.**

The Mars Rover operates in an environment with the following characteristics:

- **Partially observable:** It cannot observe every detail of its surroundings.
- **Stochastic:** The outcomes of actions may be uncertain.
- **Sequential:** Current actions affect future actions.
- **Dynamic:** Environmental conditions may change.
- **Continuous:** Position, distance and temperature can have continuous values.
- **Single-agent:** The rover performs its tasks independently.

iii. What are the actions the agent can take?

- Move forward, backward and turn.
- Capture images using cameras.
- Collect soil and rock samples.
- Analyse samples using scientific instruments.
- Avoid obstacles.
- Send collected data to Earth.
- Monitor battery power and enter safe mode when necessary.

iv. How can the performance of the agent be evaluated?

The performance of the Mars Rover can be evaluated based on:

- Accuracy of sample analysis.
- Amount of useful scientific data collected.
- Successful completion of exploration tasks.
- Efficient use of battery power.
- Safe navigation without damaging equipment.

- Successful transmission of data to Earth.

v. What is the most suitable agent architecture? Why?

A **model-based, goal-based agent architecture** is most suitable for the Mars Rover. A model-based agent maintains information about its environment using sensor data. A goalbased agent selects actions that help it achieve objectives such as exploring terrain, collecting samples and analysing rocks.

Combining these approaches allows the rover to make decisions, avoid obstacles and plan its exploration even when complete information about the environment is unavailable.

3. Eight-Queens Problem

Problem Statement: Place eight queens on an 8 × 8 chessboard so that no two queens attack each other.

Problem Components

1. **Initial State:** An empty 8 × 8 chessboard.
2. **State Space:** All possible arrangements of queens on the chessboard.
3. **Initial Configuration:** No queens are placed on the board.
4. **Actions:** Place a queen in an empty square such that it does not attack any previously placed queen.
5. **Transition Model:** Placing a queen changes the board configuration.
6. **Goal Test:** All eight queens are placed, and no two queens share the same row, column or diagonal.
7. **Path Cost:** Each queen placement can be assigned a cost of 1. A complete solution requires eight placements.

Search Strategy

Backtracking search (Depth-First Search) is suitable for solving the Eight-Queens problem.

Algorithm:

1. Start with an empty chessboard.
2. Place a queen in the first column.
3. Check whether the position is safe.
4. If safe, move to the next column and place another queen.
5. If no safe position exists, return to the previous column and move its queen to another position.
6. Repeat until all eight queens are placed safely.
7. If all columns contain a queen, the solution is found.

One Valid Solution

The following positions represent the row number of the queen in each column: **(1, 5, 8, 6, 3, 7, 2, 4)**

This configuration satisfies the row, column and diagonal constraints.

Result: The Eight-Queens problem can be solved using backtracking search by systematically exploring possible positions and rejecting unsafe configurations.

4. OLA Cab Booking Using a Problem-Solving Agent Type of Agent

A **utility-based problem-solving agent** is suitable for the OLA Cab booking application. The agent considers the customer's source, destination, preferred cab type, fare, estimated travel time and comfort. It compares available options and recommends the most suitable cab based on the customer's preferences.

Problem Components

- **Initial State:** Customer's current location and booking requirements.
- **Goal State:** Reach the destination using a suitable cab.
- **Actions:** Select a cab type, compare fares, choose a cab and confirm the booking.
- **Performance Measure:** Customer satisfaction, affordability, comfort and minimum travel time.

Pseudocode

```
BEGIN
INPUT source_location
INPUT destination_location
INPUT preferred_cab_type
INPUT budget
INPUT comfort_preference
DISPLAY available_cab_types
FOR each cab_type in available_cab_types DO
   GET estimated_fare
   GET estimated_travel_time
```

```
    GET availability
    IF cab_type is available THEN
        COMPARE fare, travel_time and comfort
    END IF
END FOR
SELECT the cab that best matches the
customer's preferences and budget
IF a suitable cab is available THEN
    DISPLAY cab_type, fare and travel_time
    ASK customer to confirm booking
    IF customer confirms THEN
        BOOK the selected cab
        DISPLAY booking confirmation
    ELSE
        DISPLAY "Booking cancelled"
    END IF
ELSE
    DISPLAY "No suitable cab available"
END IF

END
```

Explanation

The agent collects the customer's requirements, evaluates available cab options and selects the most suitable option. The customer can confirm or cancel the booking.

Result: A utility-based agent helps customers choose a suitable cab according to their budget, comfort and travel requirements.

5. Uniform Cost Search (UCS) Algorithm Problem Statement

A logistics company wants to find the least-cost path from the starting warehouse S to the destination warehouse G. Each edge represents the travel cost between two warehouses.

Given Weighted Graph

The edges and their costs are:

- S → A = 1
- S → G = 12
- A → B = 3 · A → C = 1 · B → D = 3 · C → D = 1
- C → G = 2
- D → G = 3

Algorithm

Uniform Cost Search (UCS) expands the node with the lowest cumulative path cost. It uses a priority queue to maintain the nodes in increasing order of cost. **Pseudocode**

UCS(Graph, Start, Goal)

1. Create an empty priority queue.
2. Insert Start with cost 0.
3. Create an empty explored set.

4. WHILE the priority queue is not empty DO

Remove the node with the lowest cost.

IF the node is Goal THEN
RETURN the path and total cost
END IF

IF the node is not explored THEN
Add the node to the explored set.

FOR each successor of the node DO
Calculate the new cumulative cost.

IF the successor is not explored
or a lower-cost path is found THEN
Insert or update the successor in
the priority queue.
END IF
END FOR
END IF

END WHILE

5. RETURN "No path found"

END

Step-by-Step Calculation

Path	**Calculation**	**Total Cost**
S → A	1	1
S → A → C	1 + 1	2
S → A → C → D	1 + 1 + 1	3
S → A → C → G	1 + 1 + 2	4
S → A → C → D → G	1 + 1 + 1 + 3	6
S → A → B → D → G	1 + 3 + 3 + 3	10
S → G	12	12

UCS selects the goal when it is removed from the priority queue with the lowest cumulative cost.

Optimal Path

S → A → C → G

Total Cost = 1 + 1 + 2 = **4**

Result

Uniform Cost Search finds the least-cost path from warehouse S to warehouse G with a minimum total travel cost of **4 units**. It guarantees an optimal solution when all edge costs are non-negative.